TGS-2024043854 · DAY 1 · LAB 4 · 60 MIN

Multi-Agent Video Studio — Research → Script → Production → Audit → Publish

Build a Human-AI Workforce with Autonomous AI Agents · Multi-agent (ordered pipeline) · LO2

> Build a five-agent production team on the Kanban board, render a real 25-second vertical video with HyperFrames, let an auditor agent check it, and approve it yourself before it publishes.

The business scenario

The campaign needs a 25-second TikTok/Reels launch video. One agent doing everything is slow, expensive and hard to review, so you staff a studio: a researcher finds the hook, a scriptwriter writes it, a producer renders it, an auditor checks it against the brand and compliance rules, and a publisher prepares and posts it — but only after you approve the cut.

You will deliver

- 5 Hermes profiles with routing descriptions (researcher, scriptwriter, producer, auditor, publisher)
- A Kanban task graph with dependencies and a human gate before publishing
- video-studio/research.md, script.md (scene list), output/final.mp4 (9:16)
- video-studio/audit-report.md — PASS/FAIL on V1–V9 and C1–C10
- video-studio/publish-package.md, reviewed by you, then the video posted once to your Telegram channel

Agents: 5 Hermes profiles coordinated by the Kanban board
You (the human): Review the cut and the publish package, then release the posting task (or send it back with comments)
Goal: Orchestrate a multi-agent team with role specialisation, dependencies and a human approval gate, and correlate the team design with quality, cost and speed (LO2).

Before you start

- Lab 3 complete: Telegram bot connected, the gateway running (it dispatches Kanban tasks) and your channel ID from `hermes send --list telegram`
- Node.js 22+ and FFmpeg (installed with Hermes) for video rendering

Lab files (data/)

File	What it is
`AGENTS.md`	Company context Hermes loads automatically
`brand-guide.md`	Brand voice, colours, platform formats

campaign-brief.md	Gula Melaka Cold Brew campaign brief (offer, KPIs, timeline)
compliance-rules.md	C1–C10 brand-safety and escalation rules
product-catalogue.csv	12 products and prices
templates/video-brief-template.md	Template
video-qa-checklist.md	V1–V9 video QA checks
skills/brand-compliance-audit/SKILL.md	Agent skill (SKILL.md)
DESIGN.md	Visual identity for HyperFrames
souls/researcher.SOUL.md	Role + rules for one studio profile
souls/scriptwriter.SOUL.md	Role + rules for one studio profile
souls/producer.SOUL.md	Role + rules for one studio profile
souls/auditor.SOUL.md	Role + rules for one studio profile
souls/publisher.SOUL.md	Role + rules for one studio profile
tasks/t1-research.md	Task brief (used with --body-file)
tasks/t2-script.md	Task brief (used with --body-file)
tasks/t3-produce.md	Task brief (used with --body-file)
tasks/t4-audit.md	Task brief (used with --body-file)
tasks/t5-package.md	Task brief (used with --body-file)
tasks/t6-post.md	Task brief (used with --body-file)
assets/brand-palette.png	Brand image asset
assets/cold-brew-product-card.png	Brand image asset
assets/kopi-kaki-logo.png	Brand image asset

All company data in this lab (Kopi Kaki Co., people, reviews, comments, metrics, competitors) is fictional and generated for training.

Steps

1. Install the HyperFrames video skill

HyperFrames (open source, no API key) renders MP4 videos from HTML + animation using headless Chrome and FFmpeg. Install the official optional skill and run its setup script once.

TERMINAL

```
hermes skills install official/creative/hyperframes --yes
bash "$(dirname "$(find ~/.hermes/skills -path '*/hyperframes/SKILL.md' | head
-1)")/scripts/setup.sh"
npx hyperframes doctor
```

✓ Expected: `npx hyperframes doctor` reports Node ≥ 22, FFmpeg and Chrome OK.

2. Create the shared studio workspace

All five agents share one folder so each stage can read the previous stage's output. Copy this lab's data — it includes DESIGN.md (the visual identity HyperFrames asks for), a SOUL.md per role and the auditor's skill. Install the skill BEFORE creating the profiles, because `--clone` copies your skills into each new profile.

TERMINAL

```
cp -Rn data/* ~/kopi-kaki/
mkdir -p ~/.hermes/skills/marketing && cp -R ~/kopi-kaki/skills/brand-compliance-audit
~/.hermes/skills/marketing/
mkdir -p ~/kopi-kaki/video-studio/output
# put YOUR channel id (from Lab 3: hermes send --list telegram) into the posting brief
CH="telegram:-100XXXXXXXXX"
perl -pi -e "s/telegram:<channel_id>/$CH/" ~/kopi-kaki/tasks/t6-post.md
cp ~/kopi-kaki/DESIGN.md ~/kopi-kaki/brand-guide.md ~/kopi-kaki/campaign-brief.md ~/kopi-
kaki/compliance-rules.md ~/kopi-kaki/video-qa-checklist.md ~/kopi-kaki/video-studio/
```

3. Staff the studio — five profiles with routing descriptions

Each profile is an isolated agent with its own memory and SOUL.md. The `--description` is the routing contract the Kanban orchestrator reads, so say exactly what the role takes in and hands off. `--clone` copies your model and keys.

TERMINAL

```
hermes profile create researcher --clone --description "Finds hooks, trends and proof points for a short video; writes research.md with cited sources"
hermes profile create scriptwriter --clone --description "Turns research.md into a 25-second script and 6-scene list in script.md"
hermes profile create producer --clone --description "Renders script.md into a 9:16 MP4 with HyperFrames; writes output/final.mp4"
hermes profile create auditor --clone --description "Audits a video and its copy against video-qa-checklist.md and compliance-rules.md; PASS or FAIL with fixes"
hermes profile create publisher --clone --description "Builds the publish package and posts an approved video to the Telegram channel"
for p in researcher scriptwriter producer auditor publisher; do cp ~/kopi-kaki/souls/$p.SOUL.md ~/.hermes/profiles/$p/SOUL.md; done
hermes profile list
```

✓ Expected: Five profiles listed (also visible on the dashboard's Profiles page).

4. Build the task graph with a human gate

Create the six tasks in order. Each `--parent` makes a task wait for the previous one. The publisher first only BUILDS the publish package (T5); the task that actually POSTS (T6) starts BLOCKED — only you can release it, after you have reviewed both the cut and the package. Copy each printed task ID into the next command (shown as <T1>, <T2> …).

TERMINAL

```
S=dir:$HOME/kopi-kaki/video-studio
hermes kanban create "Research hooks for the Gula Melaka Cold Brew launch short" --assignee researcher --workspace $S --tenant kopi-video --body-file ~/kopi-kaki/tasks/t1-research.md
hermes kanban create "Write the 25-second script and scene list" --assignee scriptwriter --parent <T1> --workspace $S --tenant kopi-video --body-file ~/kopi-kaki/tasks/t2-script.md
hermes kanban create "Render the 9:16 launch video with HyperFrames" --assignee producer --parent <T2> --workspace $S --tenant kopi-video --skill hyperframes --max-runtime 45m --body-file ~/kopi-kaki/tasks/t3-produce.md
hermes kanban create "Audit the video and copy (V1-V9, C1-C10)" --assignee auditor --parent <T3> --workspace $S --tenant kopi-video --skill brand-compliance-audit --body-file ~/kopi-kaki/tasks/t4-audit.md
hermes kanban create "Build the publish package (do not post)" --assignee publisher --parent <T4> --workspace $S --tenant kopi-video --body-file ~/kopi-kaki/tasks/t5-package.md
hermes kanban create "Post the approved video to the channel" --assignee publisher --parent <T5> --workspace $S --tenant kopi-video --initial-status blocked --body-file ~/kopi-kaki/tasks/t6-post.md
```

✓ Expected: `hermes kanban list --tenant kopi-video` shows 6 tasks; T6 is blocked.

5. Watch the studio work

The gateway from Lab 3 dispatches ready tasks to the right profile. Watch the lanes move (todo → ready → running → done) in the terminal or on the dashboard Kanban page. Open each task to read its comments and outputs.

TERMINAL

```
hermes kanban watch --tenant kopi-video
# or open the web dashboard:
hermes dashboard
```

✓ Expected: research.md → script.md → output/final.mp4 → audit-report.md → publish-package.md appear in ~/kopi-kaki/video-studio.

6. If the auditor says FAIL — rework, don't restart

A FAIL is the system working. The auditor's brief tells it to create a rework card for the producer and a re-audit card for itself. If it did not, add them yourself — the fix list comes from audit-report.md. The package and posting tasks wait until a PASS.

TERMINAL

```
hermes kanban create "Rework the video: apply the audit-report.md fixes" --assignee producer --
parent <T4> --workspace $S --tenant kopi-video --skill hyperframes --body "Fix V5: show SHIOK20
terms on the end card. Fix C9: AI voice-over disclosure. Re-render output/final.mp4."
hermes kanban create "Re-audit the reworked video" --assignee auditor --parent <rework_id> --
workspace $S --tenant kopi-video --skill brand-compliance-audit --body-file ~/kopi-kaki/tasks/t4-
audit.md
```

7. Human gate — review the cut AND the package, then release

Open output/final.mp4 and publish-package.md. Check them against the brief yourself: does the hook land in 2 seconds, are the price and SHIOK20 terms right, is the AI disclosure in the description, are the hashtags on-brand? If yes, unblock T6. If not, comment on T5 (or create a rework card) and keep T6 blocked.

TERMINAL

```
open ~/kopi-kaki/video-studio/output/final.mp4     # Windows/WSL: explorer.exe, Linux: xdg-open
cat ~/kopi-kaki/video-studio/publish-package.md
hermes kanban unblock <T6>
```

✓ Expected: T6 moves to ready and the publisher posts once.

8. Confirm the post and close the loop

The publisher posts the video with its caption to your channel exactly once and records the result in publish-package.md. Do not post it again yourself — check the channel and the board instead.

TERMINAL

```
hermes kanban list --tenant kopi-video
hermes kanban show <T6>
```

✓ Expected: The video is in your Telegram channel; all tasks are done.

9. Reflect — why a team beats one big agent

PROMPT — PASTE INTO HERMES CHAT

```
Read the Kanban history for tenant kopi-video (hermes kanban show for each task). Report per stage: which profile ran it, how long it took, how many retries or change requests it needed, and what was handed to the next stage. Then compare this five-agent design with a single agent doing everything: quality, cost, speed, reviewability.
```

✓ Expected: A stage-by-stage table you can use in the assessment.

Verification / checkpoint

- □ `hermes profile list` shows the five roles
- □ final.mp4 is 1080×1920, 20–30 s (`ffprobe` or the auditor's report)
- □ audit-report.md is PASS (or shows a FAIL that was fixed via request-changes)
- □ The posting task (T6) completed only after your `unblock`, and the video was posted once
- □ PP evidence: take ONE screenshot now of the Kanban board showing the tasks, dependencies and the posting gate — Task 2 of the Practical Performance uses it

Checkpoint: final.mp4 is 9:16 and 20–30 s, the audit is PASS, and the posting task only ran after you reviewed the package and unblocked it.

Troubleshooting

Symptom	Fix
Tasks never start	The dispatcher runs inside the gateway — `hermes gateway status`; restart with `hermes gateway restart`.
Render fails	Run `npx hyperframes doctor`; the producer needs Node ≥ 22 and FFmpeg. Allow `--max-runtime 45m`.
Stages can't see each other's files	Every task must use the same `--workspace dir:` path (not scratch/worktree).
Publisher doesn't know where to post	Check t6-post.md contains your real channel id (step 2), not <channel_id>.
Wrong profile picked a task	Sharpen the profile description: `hermes profile describe <name> "…"`.
Rendering too slow on your laptop	Ask the producer for `--quality draft`, or accept a 6-frame storyboard (PNG) for the audit.

Stretch challenge

Shortcut: install `official/creative/kanban-video-orchestrator`, give it the brief, and compare the team it designs with yours. Which roles did it add or merge?